K3s Homelab

GitOps · SSO · AI-Assisted DevOps

How one engineer built a production-grade K8s homelab and used Claude Code to cut operational toil.

K3s | ArgoCD GitOps | Authelia SSO | Claude Code

github.com/smii/k3sdevlab · kubernative.io

devoteam
AI-driven tech consulting

Who am I?

"If you can describe it clearly, Claude Code can build it."

Miguel Lopes

42 years old | **9 years @ Devoteam** | **kubernative.io**

Current role

Head of Cloud Native Platforms

~ 4 years in this role

Career @ Devoteam

Consultant → Principal Consultant

Led multiple containerisation projects: local & international

Managed Devoteam traineeship: non-IT → DevOps consulants

Expertise

20 years in open-source software

Hybrid Cloud

Automation

Containerisation

Beyond the keyboard

→ Traveling

→ Involved deance industry

→ Electronic Music production,

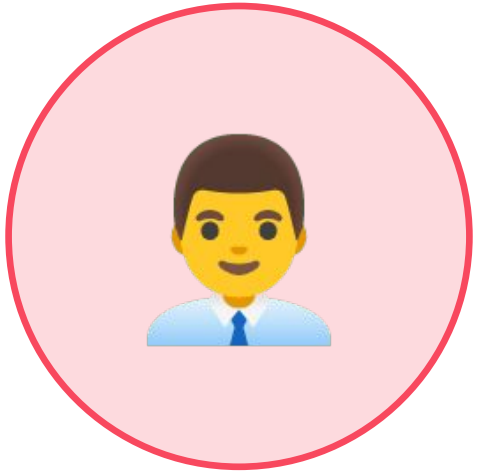
→ Reading

9y	20y	4y	∞	♪
@ Devoteam	Open-source	Tribe Manager	Futures mentored	Music producer

The why

- Want to decouple from online services and self-host. But after running 10 compose files I was done managing all manually and wanted to switch to Kubernetes for easier container management. Including a dashboard as a single pane of glass.
- Fired up Claude Code to see what kind of Kubernetes environment I could set up in limited time. I gave the AI a fun persona to work with: "A people manager who learns from his engineers during 1:1s." in 8 hours time. (Purposely left code as is without quality or efficiency check because of the persona)
- What is this? In essence, I built a portable, secure coding environment running on Kubernetes. The whole thing was built by leveraging just two AI agents(k8s specialist & SRE) and a couple of skills like an UX designer to build a `Single pane of glass` dashboard using Homepage.

Meet the builder



smii

People Manager · IT Sector

kubernative.io

GS Google Sheets Expert

Minimal traditional IT skills — but a genuine master of spreadsheets. Formulas, pivot tables, the works.

1:1 Learns in 1-on-1s

Absorbs engineering knowledge through weekly 1-on-1s with his team. Curiosity drives the conversation.

⚡ Hyperfocus mode

Lots of energy. Can suddenly lock in and go deep on a topic for hours — especially when something clicks.

🔧 Weekend lab hobbyist

Uses weekends to tinker with what engineers showed him during the week. That's how this homelab was born.

" I don't write the code — I write the CLAUDE.md. Then I watch it happen. "

The discovery

BEFORE

The weekend tinkerer approach

- Hours debugging one YAML at a time
- Googling errors, copy-pasting fixes
- SSO broken for weeks — email loop
- Each change = risk of breaking something
- No systematic test to validate anything

The Realisation

Agentic AI can do this for me

AFTER — 1 DAY

The Claude Code approach

- Write a CLAUDE.md plan in plain text
- Claude reads the whole repo in context
- Executes 8 tasks in the right order
- SSO fixed, 14 users, all ingresses patched
- Test script validates every persona

1 day	~0	8 tasks	1 file
end-to-end	manual commands	auto-executed	CLAUDE.md

Why build a homelab?

No safe sandbox

Cloud clusters cost money to experiment on. Production is not the place to learn GitOps.

SSO everywhere or nowhere

Setting up Authelia, TOTP, and forward-auth felt impossible without a real environment.

Manual ops toil

Every config change meant hours of YAML, troubleshooting, and copy-paste errors.

✅ **Solution: a fully automated, GitOps-managed K3s homelab — managed by Claude Code**

What is k3sdevlab?

A single-node K3s Kubernetes cluster running on bare metal, managed entirely through ArgoCD GitOps.

Every application is declared in Git. ArgoCD watches the repo and syncs automatically. No kubectl apply by hand.

🖥️ **Hardware**

AMD 5700 · 32 GB RAM · 500 GB SSD · Ubuntu · dnsmasq · kubernative.io

ArgoCD Deployment Waves

Wave 0	Sealed Secrets
Wave 1	Traefik · CrowdSec
Wave 2	Authelia · Tekton Pipelines
Wave 3	Gitea · Harbor · Prometheus · Grafana · Tekton Dashboard
Wave 4	Jupyter · Uptime Kuma · Hugo Blog · Sample App

🔗 argocd.rtm.kubernative.io · git.rtm.kubernative.io · grafana.rtm.kubernative.io · authelia.rtm.kubernative.io

Architecture — GitOps flow

Developer commits YAML → GitHub Repo → ArgoCD detects drift → Auto-sync applied → Cluster matches Git

Inside the K3s cluster

Ingress Layer

Traefik + cert-manager + CrowdSec

Auth Layer

Authelia SSO · TOTP · Forward-auth middleware

Apps Layer

Gitea · Harbor · Jupyter · Hugo Blog · Sample App

Observability

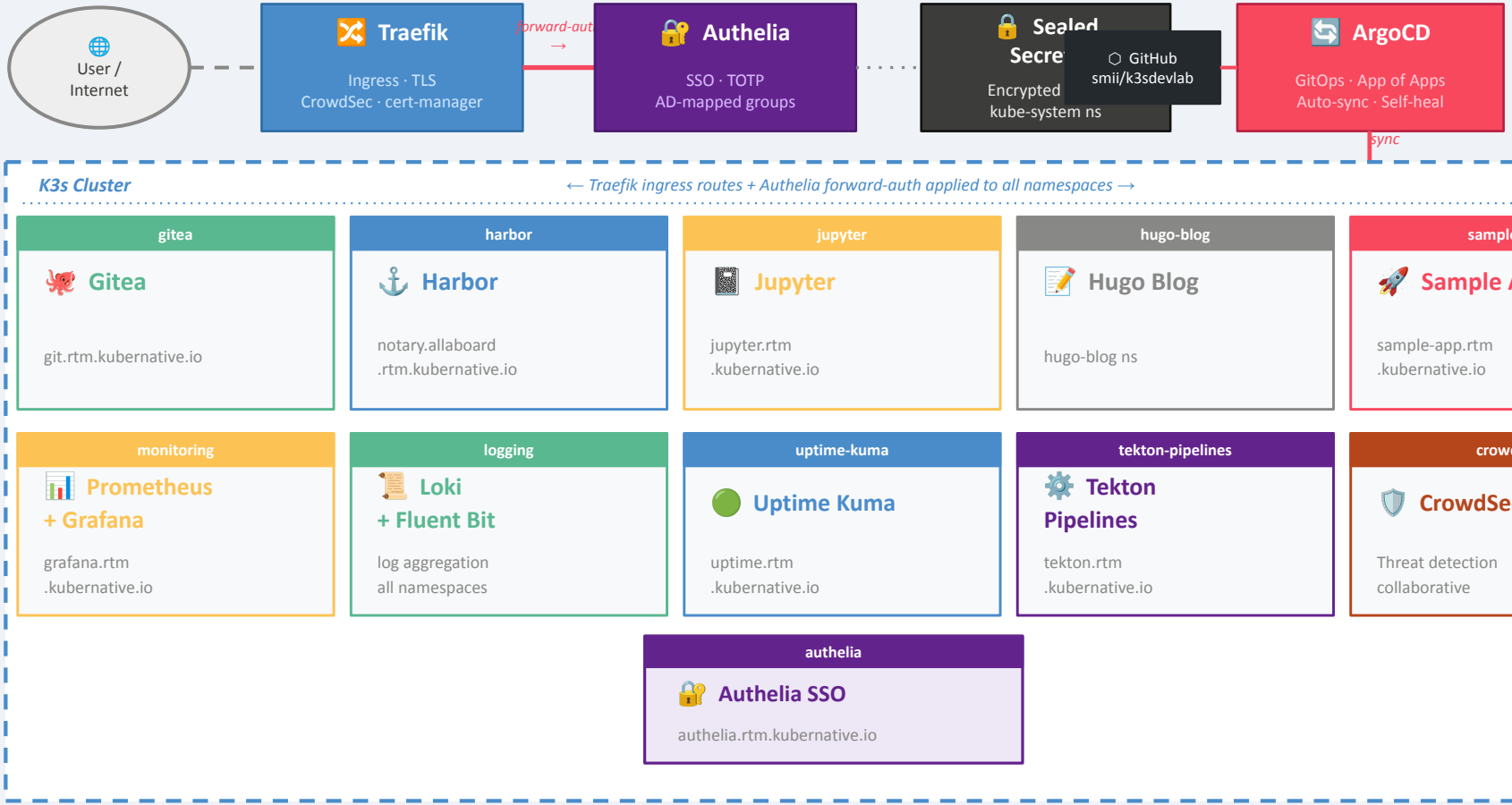
Prometheus · Grafana · Loki · Fluent Bit · Uptime Kuma

CI/CD Layer

Tekton Pipelines · Tekton Dashboard · Kaniko

k3sdevlab · Complete Architecture

kubernative.io · 192.168.1.56



Ingress / Networking · Auth / Security · GitOps / CI-CD · Dev Tools · Observability · Threat Detection

Everything in the lab

Infrastructure	Observability	Security / Auth	Dev Tools
K3s (Kubernetes)	Prometheus + Grafana	Authelia SSO	Gitea (Git)
Traefik (Ingress)	Loki + Fluent Bit	TOTP 2FA only	Harbor (Registry)
Sealed Secrets	Uptime Kuma	Forward-auth middleware	Tekton CI/CD
cert-manager	Tekton Dashboard	AD-mapped groups	Jupyter Notebook
CrowdSec	Custom dashboards	14 test personas	Hugo Blog + Sample App

GitOps — Git is the single source of truth

📋 Declarative

All config lives in YAML in Git. No manual kubectl apply ever.

🔄 Automated sync

ArgoCD auto-prune + self-heal keeps cluster aligned with Git.

🌳 App of Apps

One root ArgoCD app deploys all others — one commit to rule them all.

♻️ Reproducible

Fresh cluster? Run installer.sh — ArgoCD restores everything from Git.

83 commits on develop branch · 14 apps managed by ArgoCD · 5 deployment waves · Auto-sync + self-heal enabled

Authelia SSO — what went wrong & how it was fixed

❌ The blockers

- First login required email verification — no SMTP in homelab
- 2FA setup screen never appeared (blocked by email loop)
- Authelia v4.38+ added identity-verify step before TOTP enroll
- Notification log path broken: YAML quoting bug ('/tmp/...')
- SQLite DB in ephemeral storage — session lost on pod restart

✅ The fixes

- Switched notifier to filesystem — no email ever needed
- Disabled password reset & WebAuthn, TOTP only
- Pre-registered TOTP for all 14 users via authelia CLI
- Fixed YAML quoting: filename: /tmp/authelia-notifications.log
- Enabled PVC persistence (100 Mi) for SQLite at /config/

AD-mappable group structure

Every project gets exactly 3 groups: <project>_dev · <project>_test · <project>_prd

Project	Description	_dev (1FA)	_test (1FA)	_prd (2FA ⚡)
homelab	Core infrastructure	homelab_dev	homelab_test	homelab_prd
webapp	Web applications	webapp_dev	webapp_test	webapp_prd
devops	CI/CD tooling	devops_dev	devops_test	devops_prd
platform	Observability	platform_dev	platform_test	platform_prd

Cross-cutting: admins (full access · 1FA) · viewers (read-only · 1FA) — both map directly to AD groups

_dev + _test → one_factor | **_prd → two_factor (TOTP)** | **admins + viewers → one_factor**

Claude Code — the AI co-engineer

What Claude Code did

1. **Read the repo**
   Analyzed README, installer.sh, .env, all YAML configs
2. **Root cause analysis**
   Identified 3 root causes for SSO failure (email, TOTP, YAML bug)
3. **Generated CLAUDE.md**
   8-task instruction plan with exact file paths and snippets
4. **Rewrote user database**
   14 personas across 4 projects with argon2id hashes
5. **Patched all ingresses**
   Authelia middleware annotation added to every chart values.yaml
6. **Created test script**
   scripts/test-sso.sh validates all personas against all domains

$ How to run Claude Code remotely

```
$ npm install -g
@anthropic-ai/claude-code
```

```
$ ssh user@192.168.1.56
```

```
$ tmux new -s claude
```

```
$ cd ~/k3sdevlab
```

```
$ claude "follow CLAUDE.md"
```

Runs remotely over SSH · tmux keeps it alive if connection drops

Operational time: before vs after

~8h
Manual SSO setup time

<30m
Claude Code plan + execution

14
Test personas generated

8
Ingresses patched

Task	Manual	With Claude Code
Root-cause SSO failures	~4 hrs debugging	5 min analysis
Generate 14 users + hashes	~1 hr scripting	Automated
Patch all ingress values.yaml	~2 hrs of YAML editing	Single task
Write SSO test script	~2 hrs	Generated

The CLAUDE.md pattern

🔍 Read the Repo

Claude reads README, env files, YAML configs, installer scripts — builds full context before touching anything.

📋 CLAUDE.md Plan

A structured instruction file with 8 tasks, exact file paths, YAML snippets, and an execution order. Claude follows it step by step.

⚡ Execute Tasks

Claude modifies YAML, creates scripts, patches chart values, generates hashed passwords — all in the correct order.

✅ Verify & Test

scripts/test-sso.sh validates every persona against every domain. Pass/fail table output. Exits 1 on any failure.

💡 Key insight: Claude doesn't guess. It reads your actual config files, identifies root causes, writes a structured plan, and executes it with exact file-level precision. Same quality as a senior DevOps engineer — in minutes.

Full CI/CD loop in the homelab

1
git push to Gitea

2
Tekton triggers

3
Kaniko build

4
Push to Harbor

5
ArgoCD deploys

Sample Go App

HTTP server with Prometheus /metrics endpoint. Built with kaniko (no Docker daemon). Deployed via ArgoCD from apps/sample-app/k8s/.

Grafana Dashboards

Two auto-loaded dashboards via ConfigMap: request rate, error rate, p50/p95/p99 latency, Tekton run count + success rate.

SSO Protected

Tekton Dashboard at tekton.rtm.kubernative.io. Authelia forward-auth enforced. 2FA required for devops_prd group.

What this proves

Production practices at home

GitOps, SSO, 2FA, CI/CD, monitoring — all on bare metal, all automated.

AI agents reduce toil drastically

Claude Code turned days of debugging into a structured 30-minute plan + execution.

CLAUDE.md is the interface

Writing a clear plan file lets AI agents work autonomously and safely on real infra.

Groups → AD mapping from day 1

[project]_dev/_test/_prd naming means zero rework when connecting an external directory.

What's next

- Connect real Active Directory (LDAP)
- Migrate Authelia backend to LDAP
- Add more projects with per-project rules
- Automate TOTP re-registration workflow
- Expand Tekton pipelines for all apps

Takeaways

- Changed my perspective on AI. Coming from a semi-skeptic 6 months ago when i did a weekend run but comparing to now I see now the absolute potential. This is the future.
- Before going into automation using agents, be sure you have good understanding of all the aspects of engineering. And the tools/services you want to automate. Without that by default you will become a flaw.
- Put your pride aside ‘because you know better, or love the terminal to much. Stay updated with technology. This will be the future.

What I have learned

- AI is changing fast, we need to adept.
- Context switching
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